Desktop Application - Sandstone Width Computation

Introduction

This document serves as a tutorial for the Sandstone Width Calculation desktop application. The application has been developed using the Python *Tkinter* library. Users can run it through Anaconda, Spyder, or any other suitable Python environment to calculate the width of sandstones. The application has been tested on Windows-based operating systems, including Windows 7, 8, 10, and 11.

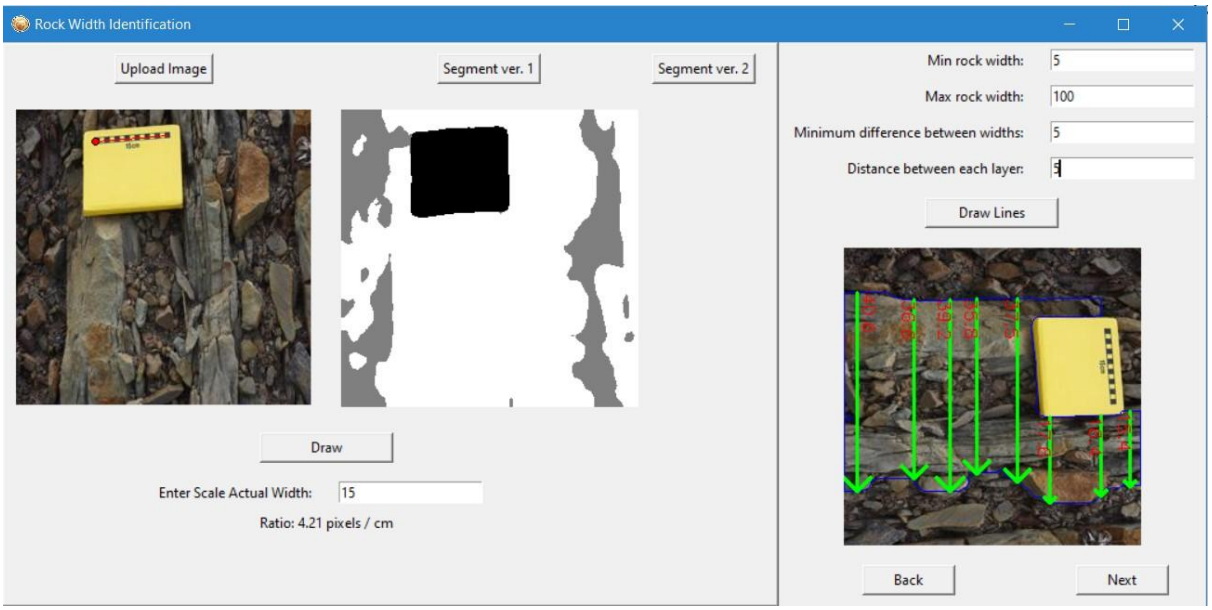


Figure 1 – A sample of user interface for the application

Expected Input

The input file should be an RGB image (.jpg, .jpeg, or .png) representing sedimentary rocks. The image must be rotated 90 degrees counterclockwise before being used in the application.

Figure 2 – A sample input file

Usage

After starting the application, the main interface will appear as shown below:

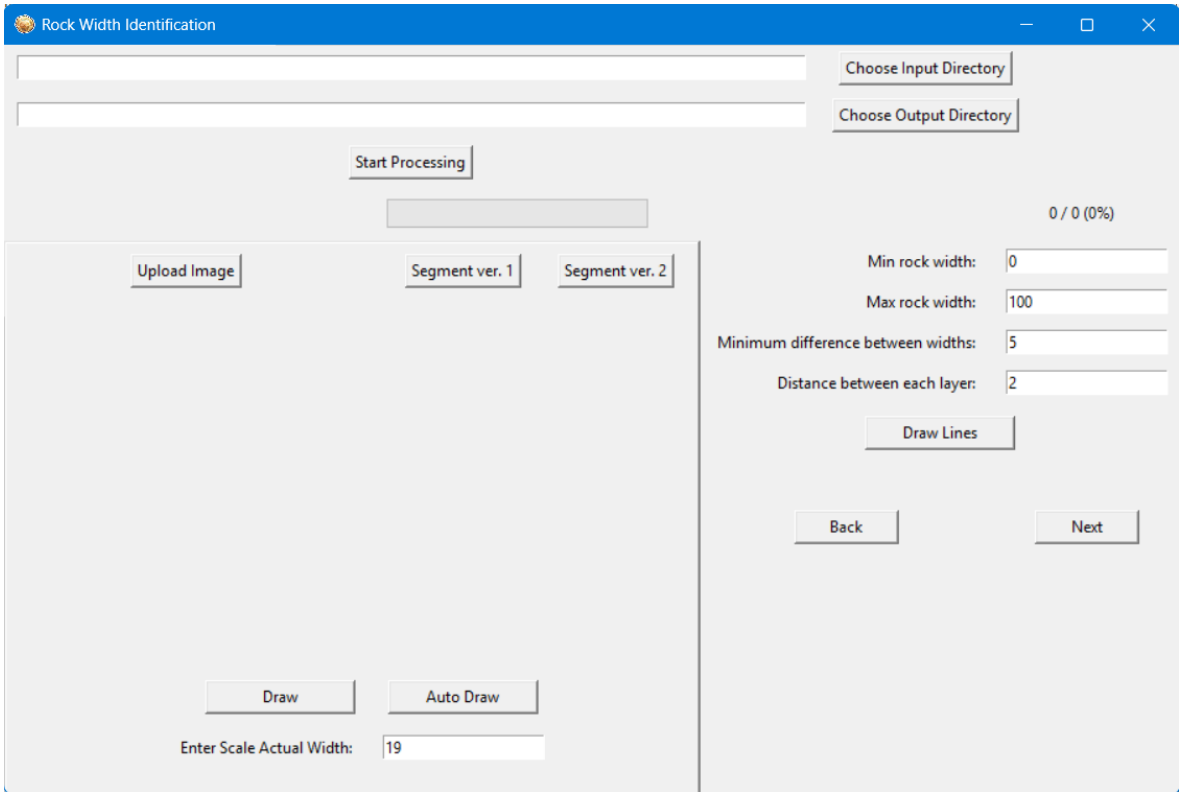


Figure 3 – User interface for the application

There are two ways to use the interface: (1) Bulk Images Mode and (2) Individual Image Mode. In Bulk Images Mode, the user is required to specify an input folder containing all the images, as well as an output folder where the segmentation results will be saved. In Individual Image Mode, the user can upload one image at a time, and the segmented result will be displayed on the screen.

The next section describes how to use the Individual Image Mode.

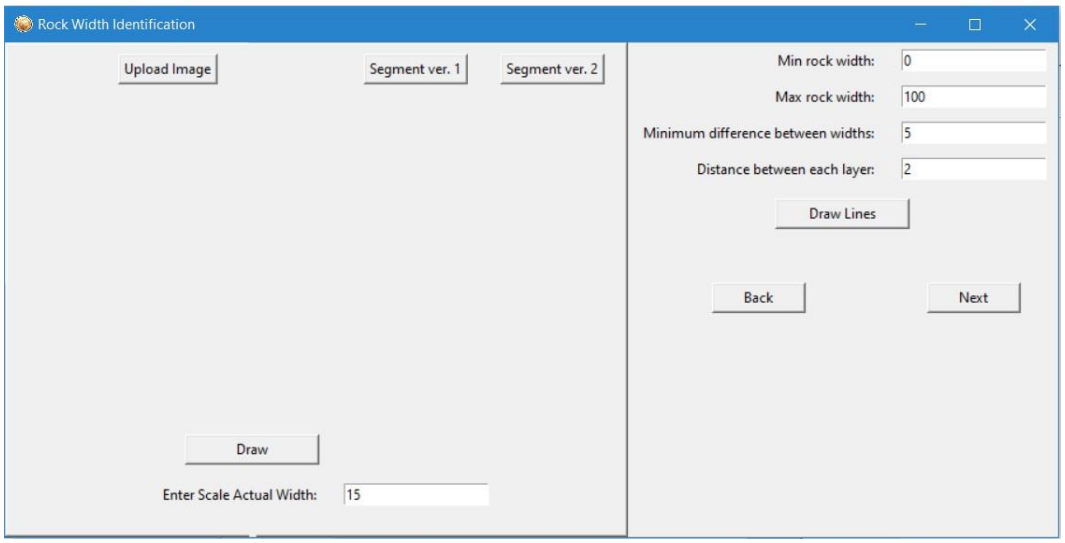


Figure 4 – User interface for the individual image mode

Step 1: Upload Image

1. Click Upload Image'
2. Pick an image of your choice, as shown below. For example, pick image '22. It will be displayed on the application:

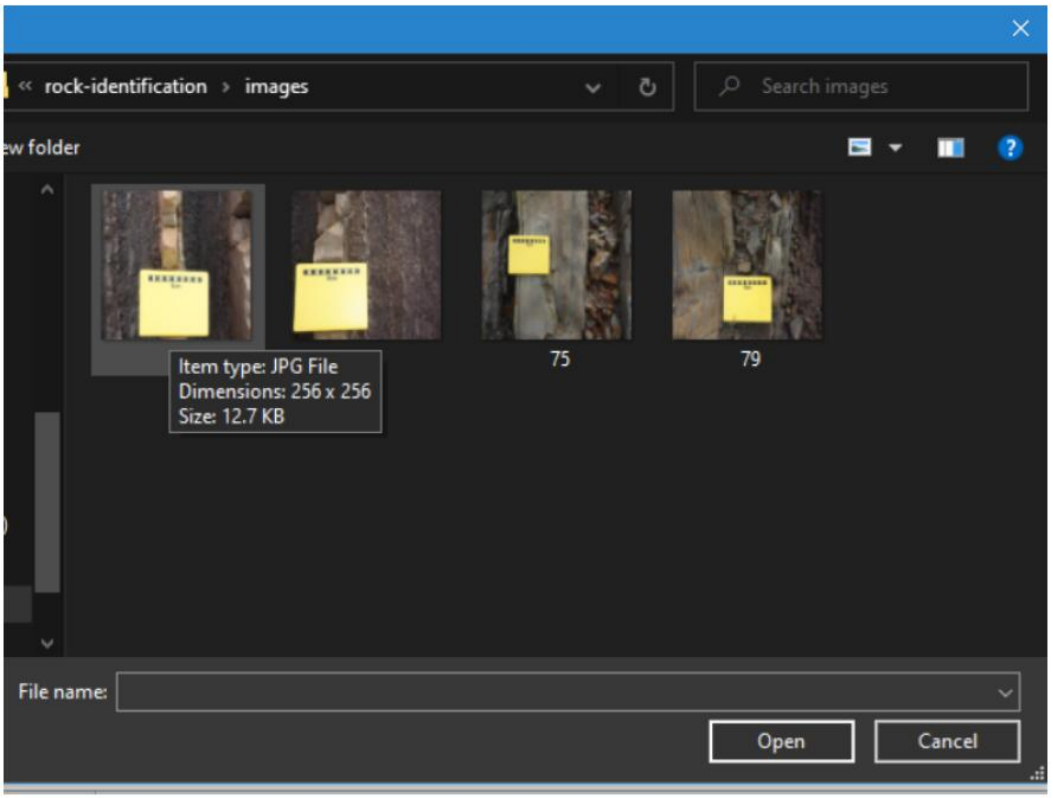


Figure 5 – Input image selection from a folder

As shown above, the image to be uploaded is a JPG file of shape 256 x 256.

3. Click 'Segment ver.1' or 'Segment ver.2' to obtain a segmentation mask of the provided image. The difference between the two is that the former uses a smaller variant of a neural network segmentation model, whereas the latter uses a larger variant.

Figure 6 – Input image selection from a folder

As shown above, the image on the left is the original image uploaded by the user, while the image on the right is the corresponding grayscale segmentation mask. In this mask:

- Black represents the background
- Gray represents mudstone
- White represents sandstone

Step 2: Calculate the Ratio

Before calculating the width of the sandstones, it is necessary to determine the **"pixel-to-actual width" ratio**. This ratio represents the number of pixels corresponding to the horizontal width of the scale in the image relative to its actual width.

1. Enter the actual width of the scale in centimeters. By default, this value is set to 15 cm.
2. Click the **"Draw"** button to draw a line over the scale. First, click on one edge of the scale to set the starting point (a red circle will appear). Then, click on the opposite edge to define the endpoint.
3. The calculated ratio will be displayed below.

Figure 7 – Scale drawing (shown in red) to determine the "pixel-to-actual width" ratio

The text *"Ratio: 6.02 pixels/cm"* is displayed, indicating that approximately 6 pixels in the image correspond to one centimeter in real-world measurement.

Step 3: Draw Lines

1. Look at the right-hand side of the application, where you will find several input fields. These fields control the lengths of the layers drawn over the sandstone segments. It is recommended to keep the default values. Then, click the **"Draw Lines"** button.

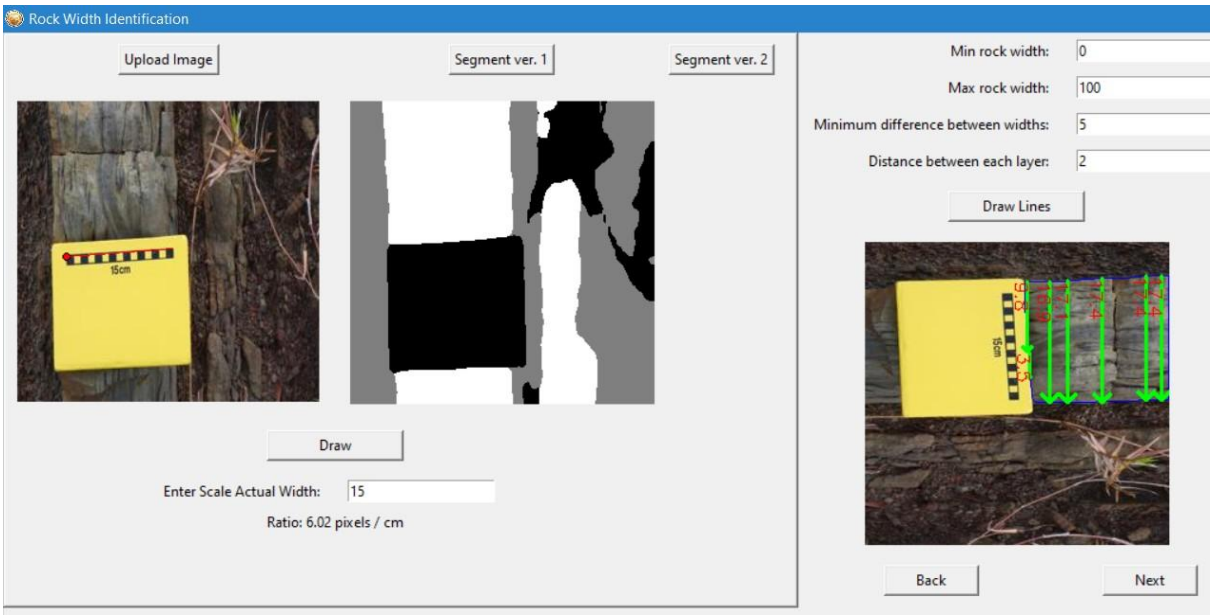


Figure 8 – Application output showing sandstone boundary detection, measurement lines, and computed width values based on the pixel-to-centimeter ratio

The following can be observed in the displayed image (Figure 8):

- The uploaded image is rotated 90 degrees clockwise.
- Blue lines are drawn around the rock to capture its boundary.
- Green lines are drawn across the shorter side of the sandstone (top to bottom).
- Red text is displayed next to each line, indicating the width of the rock in centimeters. This value is calculated by multiplying the pixel length of each green line by the pixel-to-centimeter ratio.
- Based on these measurements, the width of the rock ranges between 16.9 cm and 17.4 cm.

2. Click the **"Next"** button to display the widths of any additional rocks.

Figure 9 – Next button displays widths for additional rocks in the image

The width of the rock inside the blue delineation is between 4.7cm to 7.6cm

Multiply Width by Factor

The application resizes all images to **256 × 256** pixels. If the original image width is not 256 pixels, the calculated width should be adjusted by multiplying it by the following factor:

$$\frac{original\ width}{256}$$

Parameters Above the “Draw Lines” Button

Min and Max Rock Width

These parameters control the minimum and maximum lengths of the measurement lines (arrows) drawn on the sandstone segments.

For example, in Figure 9, representing the narrower sandstone with the yellow scale, the actual width ranges between 4 and 8 centimeters. Since values smaller than this range are not relevant, you can increase the **“Min Rock Width”** to filter out shorter lines.

Set the **Min Rock Width** to 4 and click the **“Draw Lines”** button again to update the measurements.

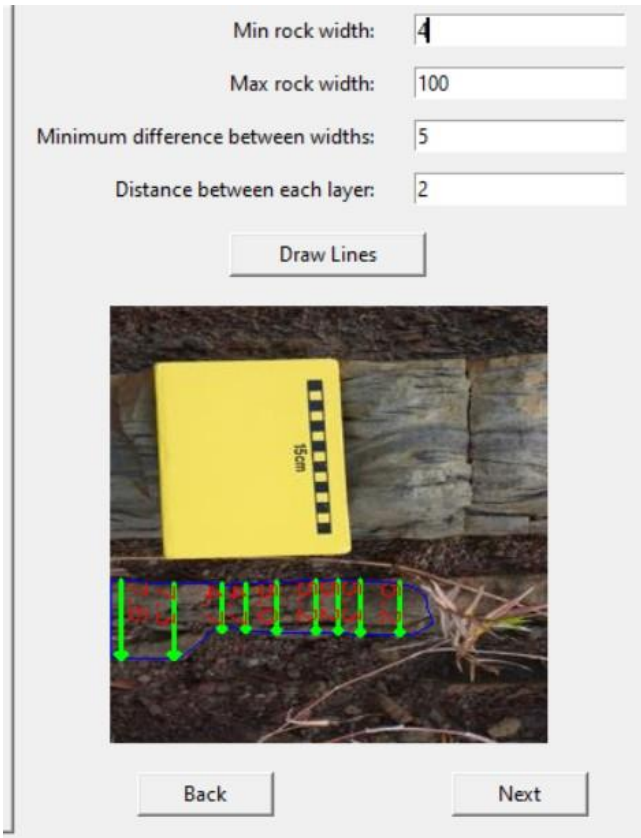


Figure 10 – Effect of minimum rock width parameter on filtering measurement lines

Care should be taken not to set the value too high; otherwise, no measurement lines will be drawn. You can also adjust the **“Max Rock Width”** parameter to prevent very long lines from being displayed.

Distance between Each Layer

This parameter controls the spacing between the measurement layers drawn over the rock. A larger value results in fewer lines being displayed, as the layers are spaced further apart.

For example, increasing the value to 3 reduces the number of lines, as shown below:

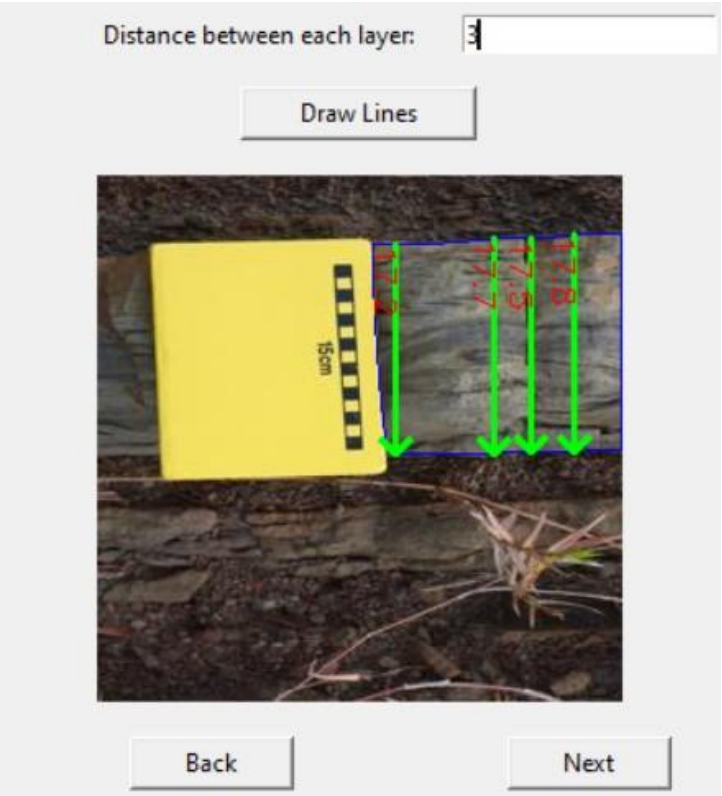


Figure 11 – Application output showing fewer measurement lines after increasing the distance between layers parameter

Minimum difference between widths:

If you want to ensure a minimum difference between the lengths of adjacent measurement lines (i.e., lines that are close to each other, defined as within twice the value of the **“Distance Between Each Layer”** parameter), you can adjust this parameter accordingly.

For example, if the value is set to 0.1:

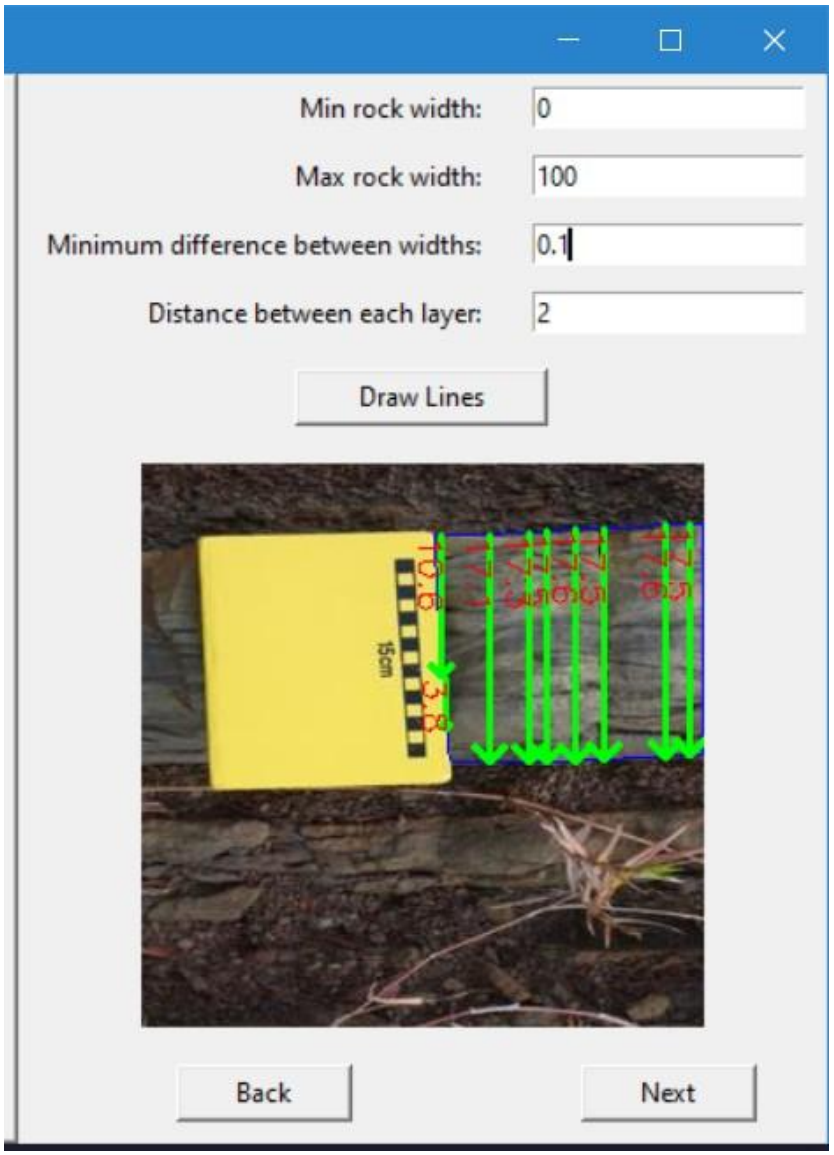


Figure 12 – Application output showing measurement lines after applying a minimum length difference constraint between adjacent layers

The arrows currently have very similar lengths. Ideally, arrows should only be placed close to each other if their lengths differ significantly. For example, two arrows should only be considered close if their lengths differ by at least **0.5 cm**:

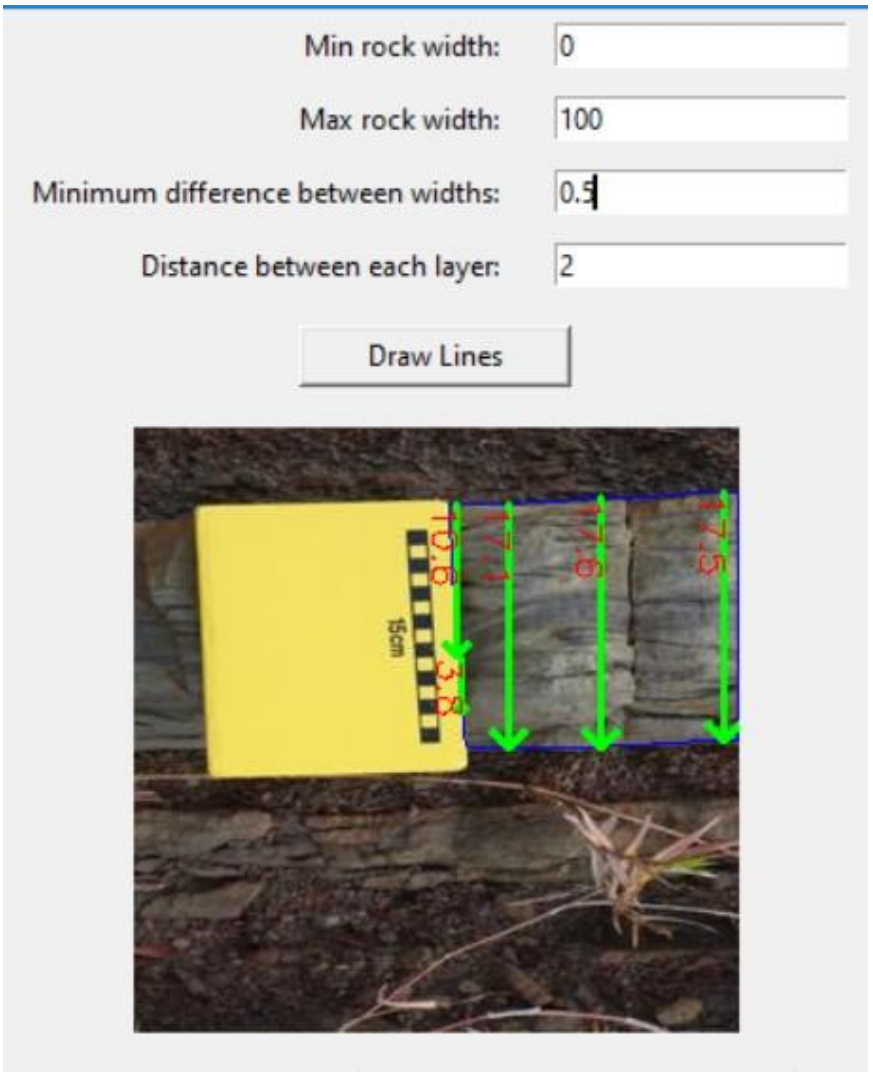


Figure 13 – Application output showing filtered measurement lines after removing arrows with similar lengths (difference < 0.5 cm)

As a result, arrows with similar lengths (i.e., a difference of less than 0.5 cm) have been removed.

A few more examples are shown below, where different values of the distance between each layer and the minimum difference between widths are used to show the effect of these parameters.

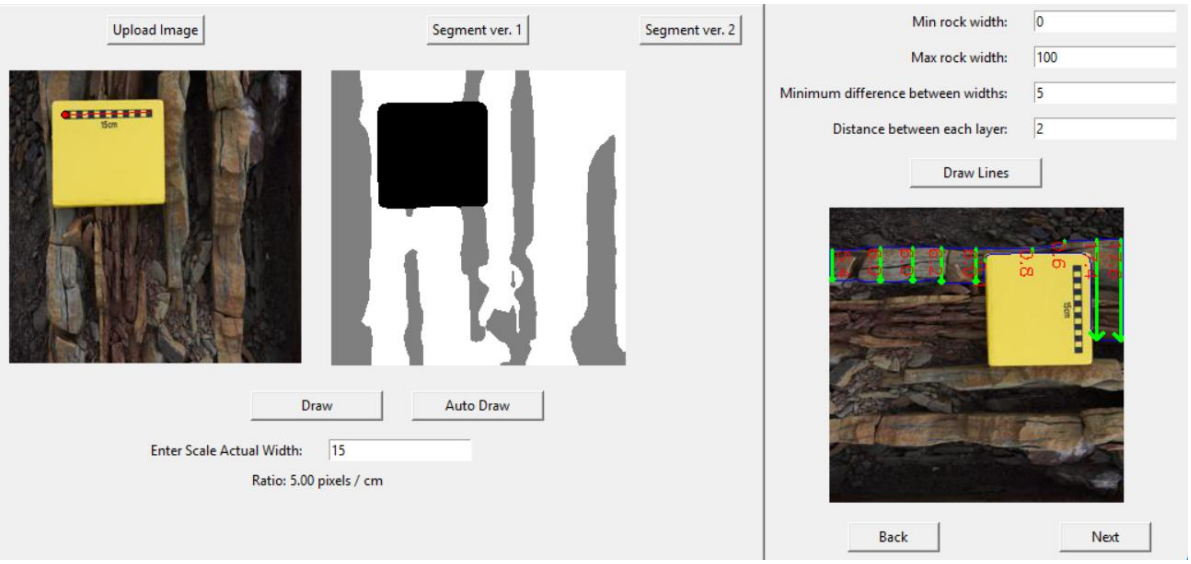


Figure 14 – Drawing the scale (shown in red line) and generating a mask and the corresponding width of the sandstone in the image

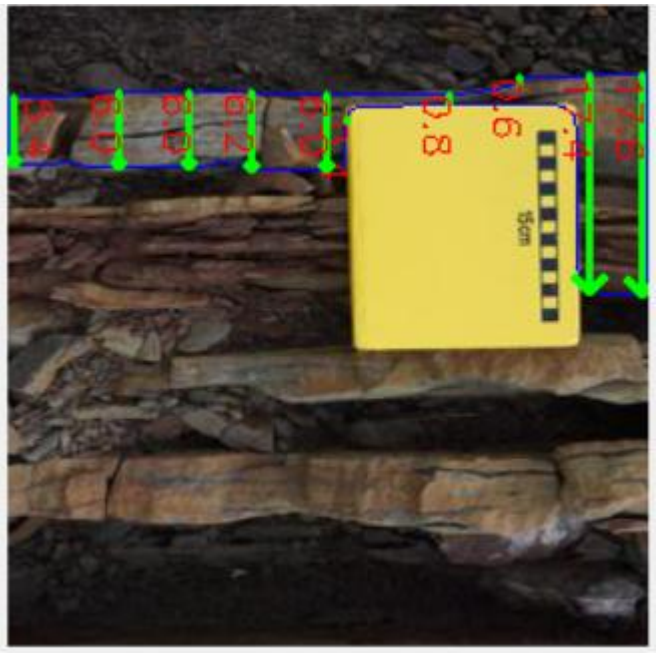


Figure 15 – distance between layers 2 cm and minimum difference between widths 5 cm

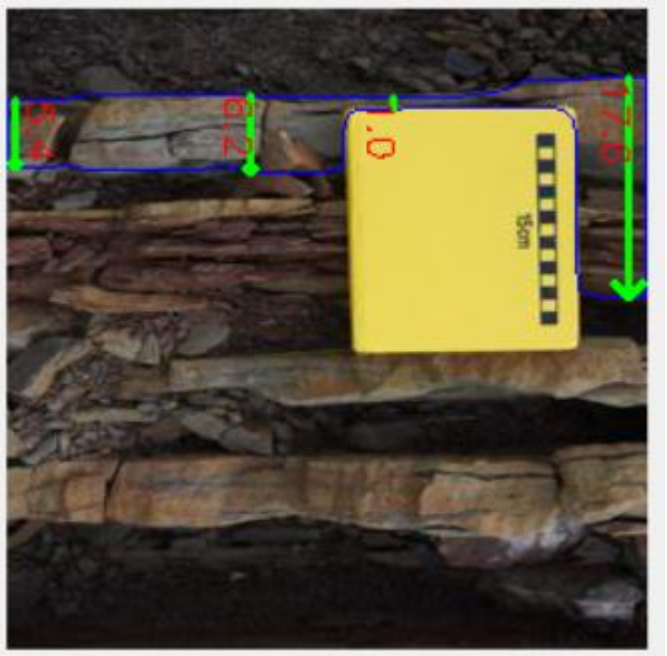


Figure 16 – distance between layers 4 cm and minimum difference between widths 5 cm

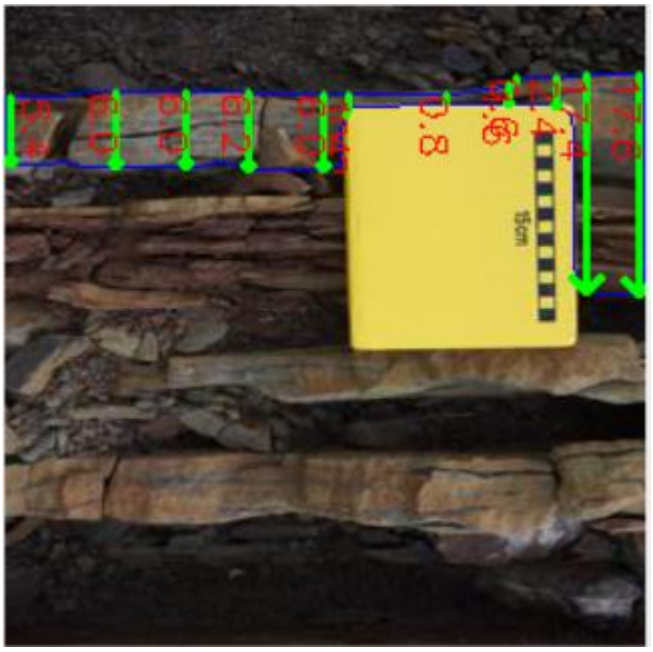


Figure 17 – distance between layers 2 cm and minimum difference between widths 0.5 cm

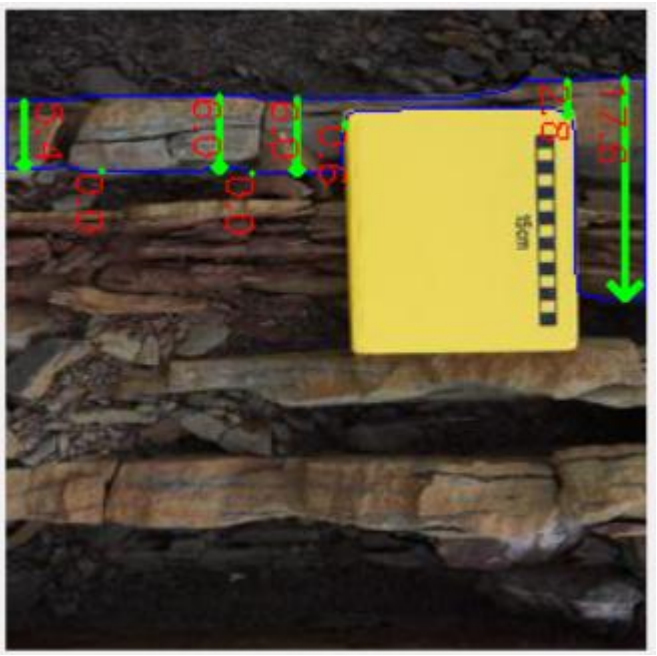


Figure 18 – distance between layers 6 cm and minimum difference between widths 5 cm

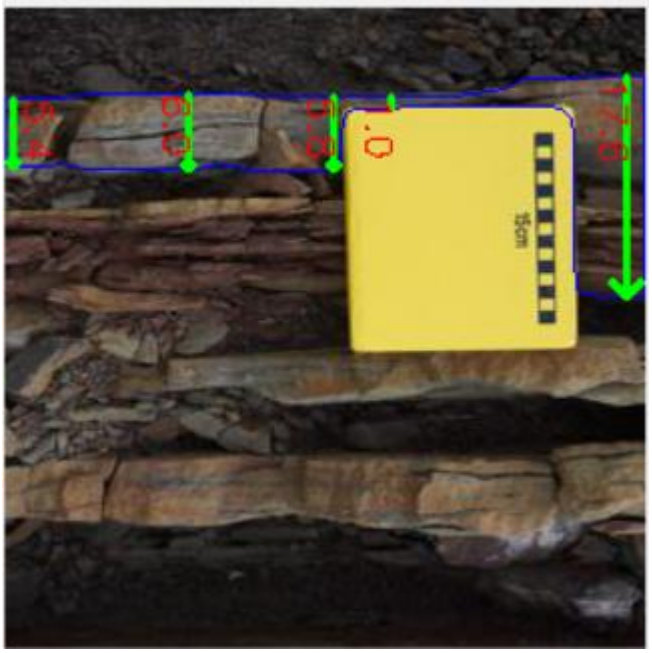
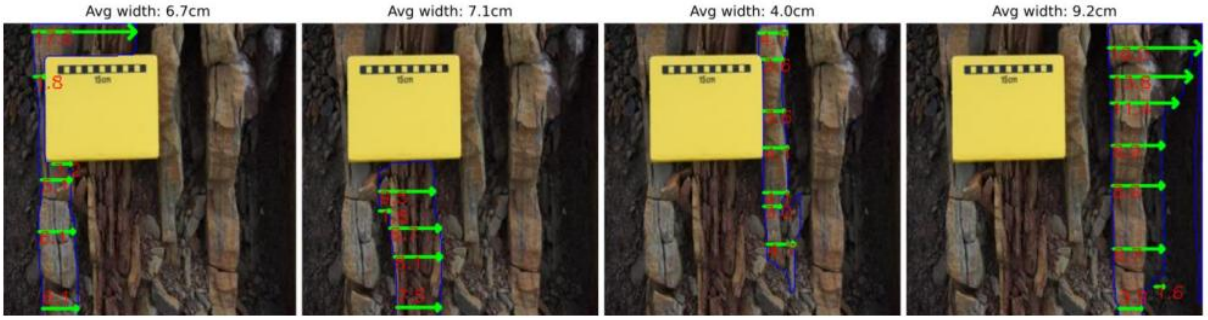
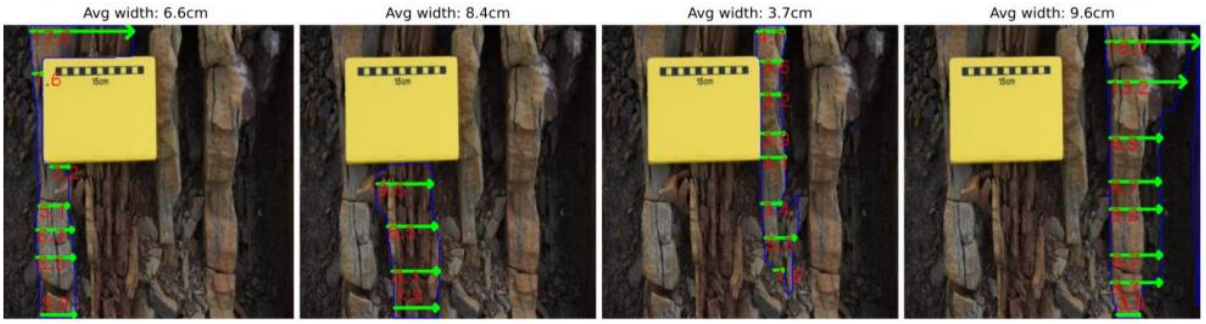


Figure 19 – distance between layers 4 cm and minimum difference between widths 2 cm

(a) Widths of sandstone with a distance of 4 cm between layers (dense measurements) and a minimum difference between widths of 0.5 cm



(b) Widths of sandstone with a distance of 6 cm between layers (fewer measurements) and a minimum difference between widths of 0.5 cm



(c) Widths of sandstone (images rotated 90° clockwise direction) with a distance of 4 cm between layers and a minimum difference between widths of 2 cm (redundant widths removed)

Figure 20 – Width measurements of multiple sandstone layers in a single image under different parameter settings - Increasing the distance between layers reduces the number of measurement lines, while increasing the minimum difference between widths removes similar measurements.